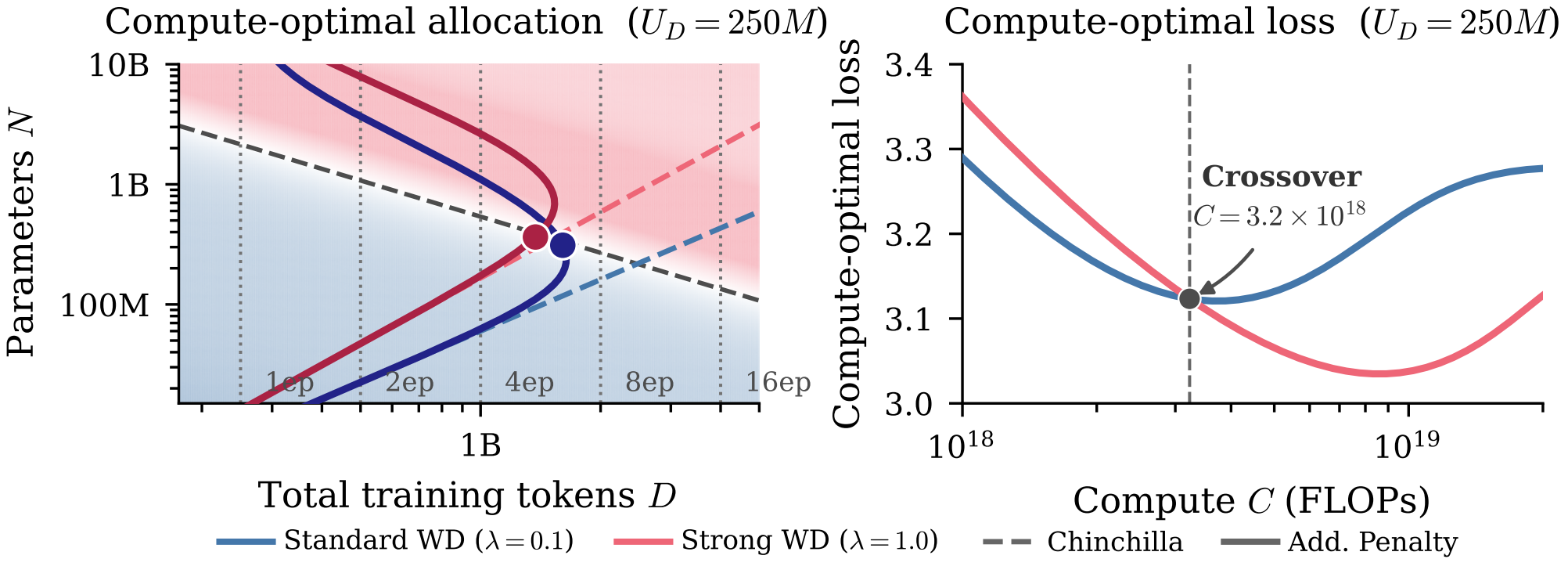

Compute-optimal allocation ($U_D = 250M$)
Parameters N
10B
1B
100M
1ep
2ep
4ep
8ep
16ep
1B
Total training tokens D
Compute-optimal loss ($U_D = 250M$)
Compute-optimal loss
3.4
3.3
3.2
3.1
3.0
Crossover
$C = 3.2 \times 10^{18}$
10^{18}
10^{19}
Compute C (FLOPs)
Standard WD ($\lambda = 0.1$)
Strong WD ($\lambda = 1.0$)
Chinchilla
Add. Penalty